Supplementary Material: Channel- and Task-Aware Object-Level Communication for Bandwidth-Constrained V2V Cooperative Perception

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I. Scope of this document

This document holds the material the main paper cites but does not print: the full cue schema, the bit-by-bit payload derivation, the complete transport sweep, the loss-position result at every rate, the complete Where2comm threshold sweep, the exploratory penalty scan, and the reproducibility apparatus. Nothing here changes a conclusion in the main paper, and the one exploratory analysis is labelled as such wherever it appears.

II. The frozen cue schema

The field list is given in the main paper's cue-schema table. Two properties hold of the schema by construction.

First, every field is ego-local: it is computed from the ego vehicle's own LiDAR sweep or its own detection output, so the vector exists before the request is issued and no field depends on a message the ego has not yet asked for.

Second, no ground truth, task metric, oracle label, delivery outcome or future information reaches the schema. The cue schema is verified against its code-defined sources, not by name matching—a wrong implementation would also rename its fields, so a name check would prove nothing.

The field of view over which the statistics are defined, the point source, and the detector checkpoint the box count depends on are all pinned in the schema manifest, because a schema frozen without pinning its inputs would be frozen against a moving input.

III. Payload derivation, bit by bit

The feature message is the encoder's bottleneck tensor. Every step below is a counted quantity, not a declared convention.

1) Elements. A dummy forward through the deployed checkpoint gives 739,200 elements passed from the encoder to the cooperative fusion stage.
2) Quantisation. Each element is transmitted at $w = 8$ bits, giving $739{,}200 \times 8 = 5{,}913{,}600$ information bits.
3) Packetisation. Payloads are cut into 8,000-bit packets, each carrying a 320-bit header: 740 packets and 236,800 header bits in total. The final packet is short and is padded, which is why the count matters.
4) Channel coding. Each packet's bits are encoded with a rate-1/2 LDPC code of information length $K = 500$ and codeword length $n = 1000$, giving $N_{\text{cw}} = 12{,}567$ codewords for the whole message.
5) Modulation. At 16-QAM, $N_{\text{cw}} \times n / \log_2 16$ channel uses give $B_F = 3.14175$ Msym per frame.

Why the codeword route rather than the direct one. Computing $(\text{info} + \text{header})/R/\log_2 M$ directly gives 3.0752 Msym. The difference from 3.14175 is codeword padding: the direct route silently assumes the information bits fill an integer number of codewords. We charge the padding, because the channel does.

The object-level message is charged through the identical chain from each frame's own box count at 184 bits per box, ranging from 0.00050 to 0.00525 Msym, i.e. 2 to 21 codewords. On the development split, the collaborator transmits an average of 24.08 boxes per frame, producing 10.10 LDPC codewords and 0.00253 Msym. The post-fusion late-detection output contains 28.40 boxes on average, but this value is not used for payload accounting. At the tightest budget tier $\beta = 0.10$ the mean object-level payload is 0.80% of the ceiling, which is why the object-level action is effectively free at every tier studied here.

IV. The operating grid

Table I gives the per-codeword erasure probability p_{cw} at every point of the operating grid: 11 SNR points crossed with the two channel types. These are the values the selector's channel cue indexes and the transport model consumes.

Two AWGN points are marked squeeze. They were absent from the link-level sweep and were filled under a preregistered table-completion rule: a point is filled only when both bracketing table entries are exactly equal, which is arithmetic on the table. This states what the table implies, not that the physical error rate is exactly zero at those points. Every other kind of gap raises rather than interpolating—AWGN falls from 0.74650 to 0.00013 between 6 and 8 dB, and a linear fill through that region would manufacture a plausible-looking number that was never measured. 2 of the 22 grid points are completed this way; the rest are measured.

TABLE I
The operating grid: per-codeword erasure probability at each SNR point and channel type. squeeze marks a point filled under the preregistered completion rule because both bracketing entries were exactly equal; it is a statement about the table, not about the channel.

	AWGN		Rayleigh	
E_s/N_0 (dB)	p_{cw}	source	p_{cw}	source
0	1.00000	measured	0.98350	measured
2	1.00000	measured	0.93850	measured
4	1.00000	measured	0.81450	measured
6	0.74650	measured	0.66050	measured
8	0.00013	measured	0.46450	measured
10	0.00000	measured	0.35050	measured
12	0.00000	measured	0.23600	measured
14	0.00000	squeeze	0.16800	measured
16	0.00000	measured	0.10450	measured
18	0.00000	squeeze	0.05950	measured
20	0.00000	measured	0.04000	measured

V. The complete transport sweep

Table II gives all 8 evaluated per-codeword loss rates in the three delivery regimes, with 4 independent channel realisations per rate averaged with equal weight. Interpolation between rates is piecewise linear in raw p, never in $\log p$: the damaged fraction of transmitted elements is approximately p, and $\log p$ diverges at $p = 0$, so it would need a special case to cover its own endpoint.

Two readings follow.

The message column measures the survival probability, not the feature action. At $p = 0.001$ a 12,567-codeword message survives with probability $q = 3.463 \times 10^{-6}$, an expectation of 0.006857 surviving frames over the whole development split. Its AP therefore sits at the ego-only value at every non-zero rate, which is a statement about message length rather than about feature-level fusion.

No monotone correction is applied. Some masks raise a frame's F_1 by removing false positives, and forcing monotonicity would use smoothing to erase a measured effect.

VI. Loss position at fixed loss amount

Table III gives the loss-position result at every rate. The construction is strict: pairs of replicates are matched on exactly equal codeword-loss counts, so the two members of a pair lose the same amount and differ only in which codewords were lost. A non-zero ΔF_1 within a matched pair therefore isolates position.

The sufficiency rule—minimum pair count, minimum frame count, and a Wilson 95% lower bound strictly above zero—was fixed before the selection ran, on a distributional argument about tie frequency rather than on a look at the outcome column.

What this supports and what it does not. It supports the statement that loss position affects task performance. It does not support the stronger statement that position matters more than amount: that comparison needs a proportion threshold fixed in advance, and none was registered. Setting one now would be setting it with the data on screen, so the stronger claim is left unadjudicated rather than argued.

VII. Where2comm: the complete threshold sweep

Table IV gives all 24 evaluated points. The communication rate is the fraction of features retained and is not converted to Msym anywhere. The evaluated implementation transmits floating-point selected features and does not execute the locked int8 quantisation, packetisation and coding chain, so a symbol count derived from it would describe a system that was never run. For the same reason the arm is excluded from bit-level budget-matched claims; it is included because it answers a different and worthwhile question—how much task performance survives feature-content selection under the same splits, field of view, ground truth and metrics.

An earlier development-split sweep of this arm predates the deterministic evaluation pipeline and is retained only as a frozen tuning record. No result reported here is derived from it.

VIII. Exploratory penalty scan

This section is an exploratory diagnostic on the development split. It was registered before it produced any number, it enters no primary criterion, and the frozen candidate was not replaced on the strength of it.

Fig. 1 and Table V scan λ over 19 points from 0.001 to 0.019. Across that interval ρ_F falls smoothly from 0.534 to 0.152, so a genuine three-action trade-off exists in the region between the preregistered grid's zero point and its smallest non-zero point. The frozen candidate's penalty $\lambda = 0.2$ lies far outside that region, at $\lambda B_F = 0.62835$ Msym against a mean net gain of 0.04890 on the rows where F is best—a ratio of 12.9.

This is a statement about the resolution of the sampled grid: no candidate on that grid selects F. It is not a measurement of whether the feature action is worth its cost.

IX. Dual payload accounting

Every payload figure in the main paper is reported under two accountings, and they are never mixed: over all frames of a split, and over only those frames where a collaborator

TABLE II
Complete transport sweep on the development split, all three delivery regimes. q is the message-level survival probability; the fragment-aware and packet columns are means over 4 realisations.

	fragment-aware		packet loss		message (all-or-nothing)	
p	AP@0.5	AP@0.7	AP@0.5	AP@0.7	q	AP@0.5
0	0.86994	0.77321	0.86994	0.77321	–	–
0.001	0.86977	0.77280	0.86952	0.77213	3.463e-06	0.72055
0.01	0.86804	0.76717	0.86619	0.76133	1.404e-55	0.72055
0.05	0.86089	0.74474	0.85208	0.72175	1.129e-280	0.72055
0.1	0.85190	0.72013	0.83503	0.68349	0	0.72055
0.25	0.82529	0.66416	0.79277	0.62354	0	0.72055
0.5	0.78452	0.61495	0.75339	0.59250	0	0.72055
0.75	0.75394	0.59346	0.73706	0.58361	0	0.72055
0.9	0.73982	0.58574	0.73315	0.58237	0	0.72055

TABLE III
Loss position at exactly equal codeword-loss counts, per rate and pooled. $\Pr(\Delta F_1 \neq 0)$ is the fraction of matched pairs whose task outcome differs.

regime	p	pairs	frames	$\Pr(\Delta F_1 \neq 0)$	Wilson 95% lo
ideal	0.001	908	789	0.0441	0.0325
ideal	0.01	311	294	0.2701	0.2238
ideal	0.05	136	131	0.5956	0.5116
ideal	0.1	103	99	0.6893	0.5945
ideal	0.25	63	63	0.8730	0.7689
ideal	0.5	57	57	0.8421	0.7264
ideal	0.75	69	69	0.8986	0.8051
ideal	0.9	84	84	0.6190	0.5122
ideal, pooled	—	1731	1201	0.2848	0.2640
packet	0.001	343	316	0.0583	0.0381
packet	0.01	69	69	0.4348	0.3243
packet	0.05	43	43	0.6279	0.4786
packet	0.1	28	28	0.6071	0.4241
packet	0.25	12	12	0.8333	0.5520
packet	0.5	26	26	0.8462	0.6647
packet	0.75	51	51	0.6078	0.4708
packet	0.9	132	132	0.1742	0.1190
packet, pooled	—	704	583	0.2557	0.2248

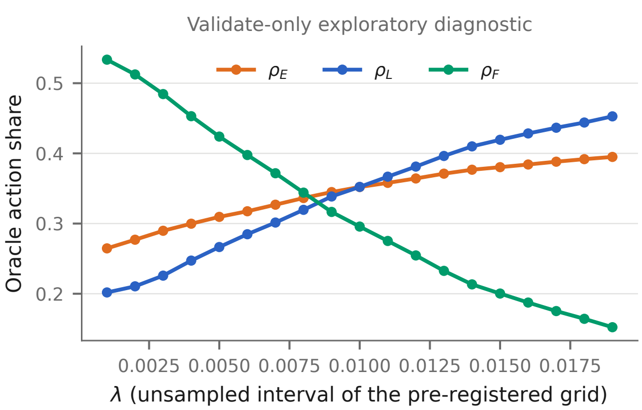

Fig. 1. Exploratory penalty scan on the development split. ρ_F falls smoothly across the interval between the preregistered grid's zero point and its smallest non-zero point. Exploratory: this figure enters no primary criterion.

is available. Frames without a collaborator cost every arm zero, so including them can only shrink absolute payloads—and could in principle inflate an apparent relative saving.

Measured, it does not. Collaborators are available on 94.52% of Test rows and 86.91% of Culver-City rows, and the relative saving is identical under both accountings on both splits. The concern was stated in advance and is settled by measurement rather than carried as a caveat.

Dual accounting is full-frame versus collaborator-available frames within CA-TOSG's single payload definition; Where2comm is reported on its native communication rate only and enters neither accounting.

X. Reproducibility

Determinism. The per-sample point shuffle is seeded on (split, scene, frame, cav). Before this fix the shuffle drew from a global unseeded generator, so two runs of the same frame could disagree—by one box on a small fraction of frames, and by 10^{-5} to 10^{-4} in AP. The fault was found by a reconstruction-consistency bridge that compares two independent runs bit for bit; after the fix that bridge reports zero differing frames and an AP difference of

TABLE IV
Where2comm, all 24 evaluated points. The rate column is the fraction of features retained, not a bit budget.

split	threshold	comm. rate	AP@0.5	AP@0.7
Culver-City	0	1.000000	0.84139	0.71694
Culver-City	0.01	0.485078	0.84118	0.71832
Culver-City	0.011	0.433140	0.84035	0.71761
Culver-City	0.012	0.382536	0.83927	0.71770
Culver-City	0.013	0.311724	0.83764	0.71664
Culver-City	0.015	0.126359	0.83419	0.71643
Culver-City	0.02	0.090778	0.83195	0.71588
Culver-City	0.025	0.072970	0.83116	0.71523
Culver-City	0.03	0.060800	0.82918	0.71432
Culver-City	0.04	0.046342	0.82578	0.71057
Culver-City	0.05	0.037854	0.82119	0.70771
Culver-City	1.1	0.000000	0.69374	0.58007
Test	0	1.000000	0.92095	0.83020
Test	0.01	0.464830	0.92004	0.83170
Test	0.011	0.422182	0.91954	0.83177
Test	0.012	0.378591	0.91936	0.83162
Test	0.013	0.313043	0.91877	0.83116
Test	0.015	0.080210	0.91808	0.83076
Test	0.02	0.052057	0.91708	0.82989
Test	0.025	0.039238	0.91511	0.82861
Test	0.03	0.031412	0.91405	0.82904
Test	0.04	0.022739	0.91121	0.82588
Test	0.05	0.017895	0.90799	0.82360
Test	1.1	0.000000	0.79757	0.68224

exactly zero. The whole seed identity set is enumerated for collisions on every run.

Bootstrap. Intervals are scene-level bootstraps: scenes are resampled with replacement and the statistic recomputed, because frames within a scene are consecutive samples of the same traffic and are far from independent. Frame-level resampling would give intervals that are too narrow by a large factor. The confirmatory unit is the scene throughout.

Selection versus evaluation. Leave-one-scene-out cross-validation on the development split is the training-time procedure that chose the candidate and the penalty; the scene-level bootstrap is the evaluation-time procedure that produced the intervals on the held-out splits. They act at different stages and neither substitutes for the other.

Products. Every number in the main paper and in this document is emitted by one generator from closed-out result files; none is typed by hand. The generator has a check mode that re-derives every macro and every table body and fails if the delivered text is not what the products produce.

TABLE V
Exploratory penalty scan, 19 points on the development split. Development split only; enters no primary criterion.

λ	oracle F_1 (scene-eq.)	payload (Msym)	ρ_E	ρ_L	ρ_F
0.001	0.88318	1.67662	0.265	0.202	0.534
0.002	0.88310	1.61050	0.277	0.211	0.512
0.003	0.88291	1.52270	0.290	0.226	0.485
0.004	0.88263	1.42373	0.300	0.247	0.453
0.005	0.88233	1.33247	0.310	0.267	0.424
0.006	0.88190	1.25009	0.318	0.285	0.398
0.007	0.88135	1.16848	0.327	0.301	0.372
0.008	0.88083	1.08177	0.336	0.320	0.344
0.009	0.88015	0.99541	0.345	0.339	0.317
0.01	0.87954	0.92989	0.352	0.352	0.296
0.011	0.87888	0.86552	0.358	0.367	0.275
0.012	0.87818	0.80072	0.364	0.381	0.255
0.013	0.87728	0.73174	0.371	0.396	0.233
0.014	0.87645	0.67126	0.377	0.410	0.213
0.015	0.87597	0.63025	0.380	0.419	0.200
0.016	0.87535	0.58995	0.384	0.428	0.187
0.017	0.87479	0.55182	0.388	0.437	0.175
0.018	0.87422	0.51715	0.392	0.444	0.164
0.019	0.87354	0.47959	0.395	0.453	0.152